Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/15/2016
PI Name:	Kucik, Dennis F. M.D., Ph.D.		
Project Title:	Radiation-Induced Early Changes in Gene and Protein Expression, Lipid Oxidative States, and Vascular Function Related to Atherosclerosis		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	35205-4831	**Congressional District:**	7
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2016 when PI Kucik retired;		
Key Personnel Changes/Previous PI:	Dr. Dennis Kucik is no longer Principal Investigator as of January 1, 2017. New Principal Investigator is Dr. Janusz Kabarowski.		
COI Name (Institution):	Kabarowski, Janusz (University of Alabama, Birmingham) White, Charles (University of Alabama, Birmingham)		
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Task Description:	Epidemiological studies have established that radiation from a number of terrestrial sources increases the risk of atherosclerosis (a cause of heart attack and stroke), even at relatively low doses. The effect of galacto-cosmic radiation (GCR), such as encountered on deep space missions, is not clear, however, because this kind of radiation interacts with tissues very differently than does most terrestrial radiation. In our previous project, we showed that heavy ions, an important component of GCR, indeed accelerate the development of atherosclerosis-related plaques in the mouse, but only at relatively high doses. Mouse models, however, are known (from comparison of animal experiments to human epidemiological data) to require much higher radiation doses for plaque formation than humans, so while this type of experiment yields valuable information about the effect of radiation, it is not very useful in predicting risk at a given dose. This is likely due to species differences in lipid metabolism and the immune system. Plaque formation is a very late endpoint, however. Very early steps in atherosclerosis, such as changes in gene expression, protein expression, and lipid oxidation, should be much more similar in humans and mice. Analysis of these early changes in response to doses of simulated GCR that astronauts are likely to encounter on a Mars mission will enable us to better predict the risk of cardiovascular consequences from space radiation while it increases our understanding of the mechanisms involved. Our hypothesis is that irradiation of blood vessels leads to early, species-independent changes in endothelial cell genes, proteins, and lipids that cause vascular dysfunction and are therefore likely to be predictive of cardiovascular disease. An understanding of these changes can be used to help predict risk and propose countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Atherosclerosis is a major adverse effect of therapeutic radiation. It is not always possible to avoid exposure of the heart and major arteries, especially for treatment of head-and-neck cancers and breast cancer. This project will provide important new information about the mechanism of radiation-induced atherosclerosis to help in designing possible mitigating strategies. Moreover, the use of ion beams for cancer treatment is becoming ever more prevalent. In the U.S., proton therapy is becoming more widespread, and carbon-ion therapy is available in Europe and Asia. Little is known about the cardiovascular effects of these new radiation treatment modalities. This work will greatly enhance our understanding of possible adverse effects.
Task Progress:	We have arranged our results from this, our second year of funding, in terms of progress on Roadmap questions below. Degen-1, “How can tissue specific risk models be developed for the major degenerative tissue risks, including heart, circulatory, endocrine, digestive, lens and other tissue systems in order to estimate GCR and SPE risks for degenerative diseases?” Our model, focusing on radiation-induced early circulatory system atherosclerotic events rather than plaque formation, is in sharp contrast to studies employing plaque formation as an endpoint, where wild type mice fed a normal diet are never used, since they never develop full-blown plaques. This combination of wild type mice on a standard diet and early changes as endpoints constituted a novel model for space-radiation-induced murine atherosclerotic changes. In this project we are investigating radiation-induced changes in specific tissues. The first tissue model we used is the aorta, an atherosclerosis-prone major artery. In this protocol, mice are whole-body irradiated at NASA Space Radiation Laboratory (NSRL), transported back to the University of Alabama at Birmingham, and then endothelial adhesiveness is assayed at 2- and 6-weeks post-irradiation. Whole aortas are removed from the mouse and immediately assayed by opening the vessel to expose the endothelium that lines the inner vessel wall, pinning the aorta to dental wax and then allowing added cultured white blood cells (WBCs) cells to adhere for 30 minutes under shear stress. Last year, we demonstrated that endothelial adhesiveness for WBCs was significantly increased at 2 weeks post-irradiation at 0.5Gy iron ion (56Fe), a space-relevant heavy-ion dose. This year, we extended the dose response curve to 0.2 Gy and 0.1 Gy 600 MeV 56Fe. We now have solid data that 0.2 Gy significantly increases adhesiveness. We are processing the 0.1 Gy data and will know soon whether this dose is also effective. This is an important result, because we now know that this early pro-atherosclerotic effect occurs at heavy ion doses well within the range to which astronauts are expected to be exposed on a Mars mission. Moreover, increased aortic endothelial adhesion is an early, necessary step in the later development of atherosclerotic plaque. Consistent with this, in mouse atherosclerosis models where key adhesion molecules are genetically deleted to interfere with this step, mice are relatively protected from plaque formation. In the more pro-atherogenic environment of human blood vessels, this change in adhesiveness would reasonably be expected to contribute to later cardiovascular disease. We have also recently performed irradiation experiments with protons. Early results suggest that 200 MeV protons at space-relevant doses do not seem to affect aortic endothelial adhesiveness. Moreover, at these doses protons do not increase carotid artery wall thickening, an independent measure of pro-atherosclerotic damage. This suggests that there may not be a significant atherosclerotic risk from space-relevant doses of protons, and that risk management and countermeasures may best be focused on the heavy-ion component of galactic cosmic radiation. These results constitute progress on our Aim #3: Determine the dose dependence of functional consequences of 56Fe and protons with respect to perturbations in endothelial reactivity and aortic adhesivity. Degen-2, “What are the mechanisms of degenerative tissues risks in the heart, circulatory, endocrine, digestive, lens, and other tissue systems? What surrogate endpoints do they suggest?” To investigate the mechanism of radiation-induced atherosclerosis in the circulatory system and to identify early markers of disease, we irradiated mice at the NSRL and collected both plasma and homogenized aorta samples at 2- and 6-weeks post-irradiation for genomic, proteomic and lipidomic analysis. So far, we have concentrated on lipid analysis and the results are striking. We had expected some alteration in plasma lipids because of their major contribution to the development of atherosclerosis. Effects on lipids might take many forms, however. For example, one common effect of tissue injury is alteration of circulating cholesterol levels. Many injurious and inflammatory stimuli result in an acute-phase-type response, including reductions in high-density lipoprotein (HDL), which is responsible for reverse cholesterol and lipid transport in the body. This cannot explain the pro-atherogenic effect of heavy ions, however, because we previously showed that exposure of mice to 56Fe radiation at doses that accelerated the development of atherosclerotic plaques did not significantly change circulating levels of HDL. Therefore, to better understand potentially pro-atherogenic effects of heavy-ion radiation, we used a

	lipidomics/metabolomics approach. We evaluated plasma lipid changes in wildtype mice six weeks following total body exposure to 2 Gy 56Fe radiation by using sequential window acquisition of all theoretical spectra (SWATH) mass spectrometry. We found that circulating lipids were mostly reduced following 56Fe exposure and that this occurred independently of changes in lipoprotein (HDL and LDL) levels, fatty acid precursor levels, or lipid oxidation. Interestingly, most of the phospholipids changed with 56Fe exposure were of a particular type that regulate a variety of physiological and pathophysiological processes. Lower-dose data (0.2 Gy and 0.5 Gy 600 MeV 56Fe) are now being analyzed. Early results indicate a dose-dependent effect. We are finding that as the absorbed dose increases, a larger number of changes are found, but significant effects occur at doses as low as 0.2 Gy. We conclude that exposure to 56Fe does not result in a sustained acute phase response to injury capable of reducing plasma HDL levels, yet leads to alterations in specific plasma lipids that could mediate systemic effects on the vasculature that are relevant to heightened atherosclerosis risk. These findings have implications for future countermeasures. This constitutes progress on Aim #2: Determine, as a function of radiation dose, changes in oxidation states of aortic endothelial cell lipids, and demonstrates that radiation results in lipidomic changes that predict a possible mechanism for disease progression. NOTE (7/13/2017): Dr. Dennis Kucik is no longer Principal Investigator as of January 1, 2017. Project continues with Principal Investigator Dr. Janusz Kabarowski, title "Radiation-Induced Early Changes in Gene and Protein Expression, Lipid Oxidative States, and Vascular Function Related to Atherosclerosis (PI: Kabarowski)." See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 04/12/2018)
Abstracts for Journals and Proceedings	Gupta K, Chanda D, Kabarowski JH, Kucik DF. "Heavy-ion Irradiation Causes an Early Atherosclerotic Change in Mice at a Mars-mission Relevant Dose." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Abstracts. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Chanda D, Gupta K, Kabarowski JH, Kucik DF. "In vivo 56Fe irradiation of wild-type C57BL/6 mice at space-relevant doses results in increased aortic endothelial adhesiveness, an early atherosclerotic change." 61st Annual Meeting of the Radiation Research Society, Weston, FL, September 19-22, 2015. Meeting proceedings. 61st Annual Meeting of the Radiation Research Society, Weston, FL, September 19-22, 2015. , Sep-2015
Abstracts for Journals and Proceedings	Kabarowski JH, Barnes S, Gupta K, Kucik DF. "56Fe alters the composition of circulating plasma lipids in wild type mice without affecting levels of HDL and LDL." 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016